

DHUM101
Socially Responsible Computing

University of Otago
Te Kura Rorohiko
School of Computing
Te Reo Inarahi me te Mātai Wetereo
English and Linguistics







Perera et al. 2026

Lecture 4
Taking responsibility for technological development

Holger Regenbrecht
Semester 2, 2026

What is that all about?

Ethics and Morality

	Ethics	Morality
What are they?	Rules of conduct recognised to particular class of human actions or group or culture.	Principles or habits with respect to right or wrong conduct. While morals also prescribe do's and don'ts, morality is ultimately a personal compass of right and wrong.
What is their source?	Social System (External)	Individual (Internal)
Why we do it?	Societies pressure saying it is the right thing to do	Personally believe something to be right or wrong

"Ethics vs Morals." *Diffen.com*. Diffen LLC, n.d. Web. 15 May 2019.< https://www.diffen.com/difference/Ethics_vs_Morals >

Information and Computer Scientists as Engineers

ABSTRACT *Most American engineers believe that they have a responsibility for the safety and well-being of society, but whence does this responsibility arise? What does it entail? After describing engineering practice in America as compared with the practice of other professions, this paper examines two standard types of accounts of the social responsibilities of professionals. While neither provides a satisfactory account of the social responsibilities of American engineers, several lessons are learned by uncovering their weaknesses. Identifying the framework in which professional rights and responsibilities are justified, I argue that an end or primary good is the starting place for conceptualizing a profession, and justifying its existence and shape. Too little attention has been paid to the end(s) of engineering. The social responsibilities of American engineers as defined in the present system of engineering are ambiguous and weak. I indicate how the case for assigning American engineers stronger social responsibilities must be made by starting with the end(s) of engineering. I argue that, at present, American engineers do not have social responsibilities as engineers, though they do have social responsibilities as persons.*

Johnson, D. G. (1992).
Do engineers have social responsibilities?.
Journal of Applied Philosophy, *9*(1), 21-34.

Information and Computer Scientists as Engineers

ABSTRACT *Most American engineers believe that they have a responsibility for the safety and well-being of society, but whence does this responsibility arise? What does it entail? After describing engineering practice in America as compared with the practice of other professions, this paper examines two standard types of accounts of the social responsibilities of professionals. While neither provides a satisfactory account of the social responsibilities of American engineers, several lessons are learned by uncovering their weaknesses. Identifying the framework in which professional rights and responsibilities are justified, I argue that an end or primary good is the starting place for conceptualizing a profession, and justifying its existence and shape. Too little attention has been paid to the end(s) of engineering. The social responsibilities of American engineers as defined in the present system of engineering are ambiguous and weak. I indicate how the case for assigning American engineers stronger social responsibilities must be made by starting with the end(s) of engineering. I argue that, at present, American engineers do not have social responsibilities as engineers, though they do have social responsibilities as persons.*

Johnson, D. G. (1992).
Do engineers have social responsibilities?.
Journal of Applied Philosophy, *9*(1), 21-34.

Information and Computer Scientists as Engineers

Responsibilities?

Responsibilities / Ethical Duties

Duties as:

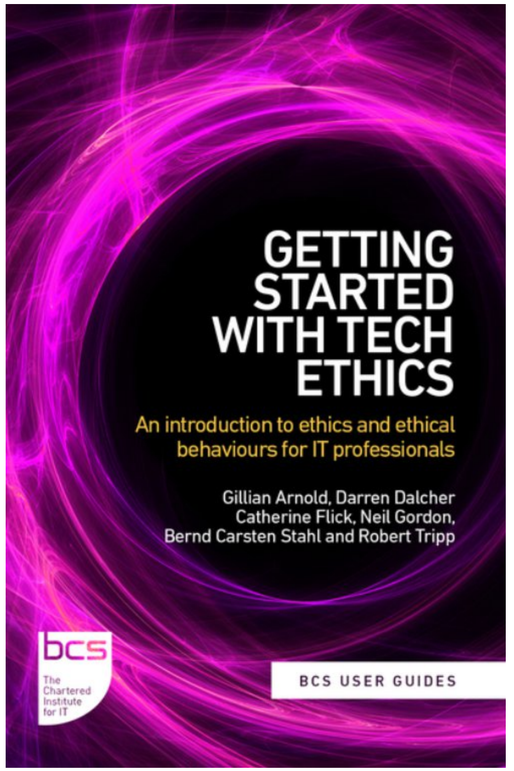
- Engineers
- Scientists
- Educators
- Consultants
- …



Johnson (1992)

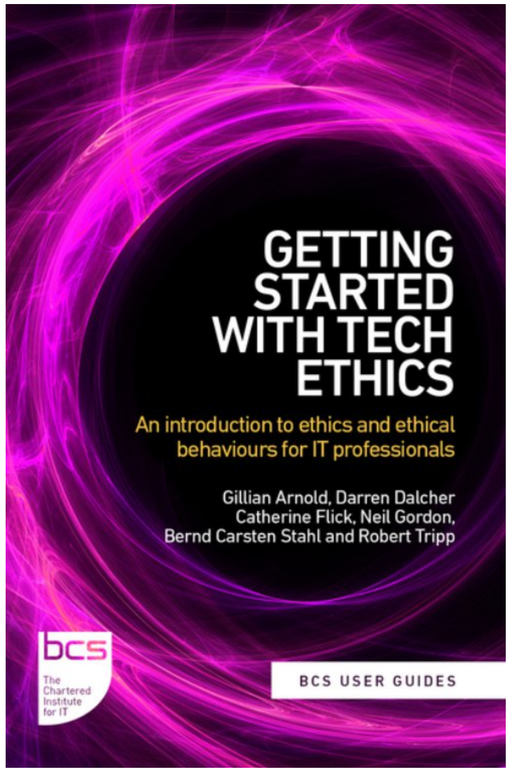
Professional Ethics: Types of Responsibility

- Causal responsibility
- Legal Responsibility
- Moral Responsibility
- Role Responsibility



Personal Ethics: What is the right thing to do?

- Is it legal?
- Is it fair?
- How will I feel afterwards?
- Can I really justify it?



Transfer of Responsibilities Assumptions (transcendent responsibility)

- Trust in ability
- Trust in competence
- Security in knowledge
- Comfort and peace of mind

'Astonishing greed' sunk Swann

Otago Daily Times

Home » News » Dunedin

Thu, 12 Mar 2009

News: Dunedin | Michael Swann

"Fraud on a grand scale" was how Justice Lyn Stevens yesterday described the actions of former Otago District Health Board chief information officer Michael Swann and his friend and business associate Kerry Harford.



Former Otago District Health Board chief information officer Michael Swann at his sentencing in the High Court at Dunedin yesterday, for defrauding the board of $16.9 million. Photo by staff photographer.

The two men were each appearing in the High Court at Dunedin for sentence after being found guilty of defrauding the board of almost $17 million.

- SDHB board members and senior staff without sufficient IT literacy
- Michael Swann IT expert
- greedy
- in powerful position
- unethical and illegal behaviour

Responsibilities / Ethical Duties

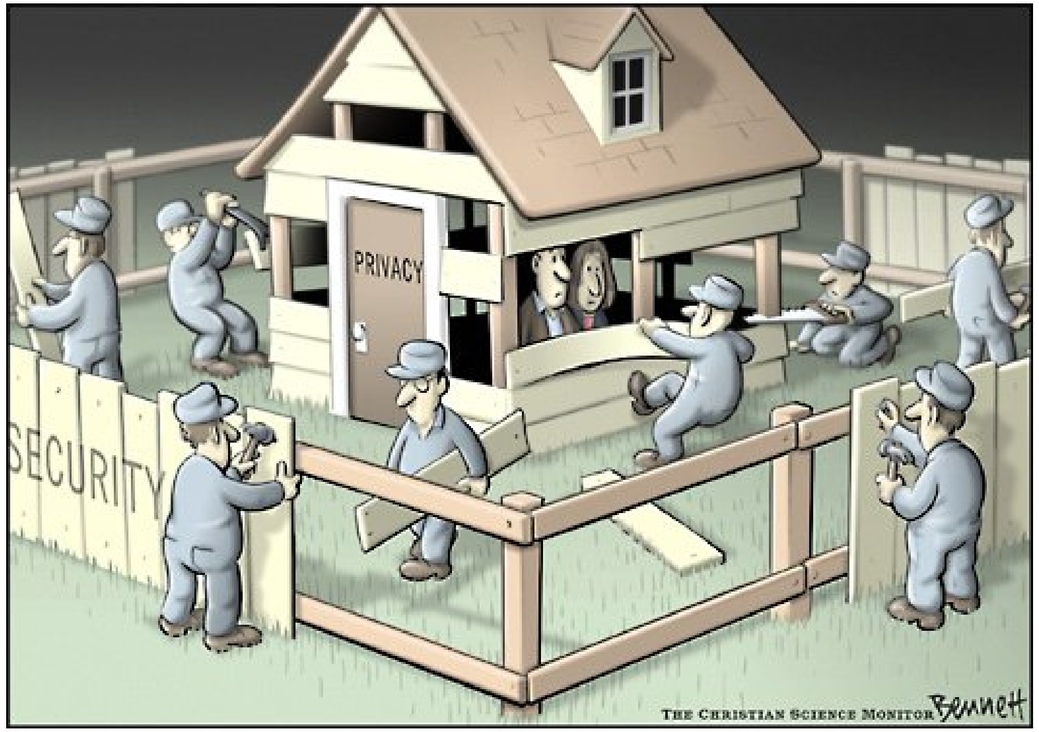


youtube Little Britain

Privacy

Privacy

- What do you consider private?
- Would you read someone else's email, word documents, spreadsheet files?
- Would you post someone else's private files?
- If you could listen to other's email or chat conversations, would you?
- What would you do if you thought your conversations were being listened to?
- Does the idea feel wrong?
- Would you tell your boss?
- Would you like to have your actions appear in the newspaper or on TV?
- Would you be proud of yourself in public for having done what you're thinking about?



New Zealand Privacy Act 2020

A quick tour of the privacy principles

The Privacy Act 2020 has 13 privacy principles that govern how you should collect, handle and use personal information.

1 You can only collect personal information if it is for a lawful purpose and the information is necessary for that purpose. You should not require identifying information if it is not necessary for your purpose.

2 You should generally collect personal information directly from the person it is about. Because that won't always be possible, you can collect it from other people in certain situations. For instance, if:

- the person concerned gives you permission
- collecting it in another way would not prejudice the person's interests
- collecting the information from the person directly would undermine the purpose of collection
- you are getting it from a publicly available source.

3 When you collect personal information, you must take reasonable steps to make sure that the person knows:

- why it's being collected
- who will receive it
- whether giving it is compulsory or voluntary
- what will happen if they don't give you the information.

Sometimes there may be good reasons for not letting a person know you are collecting their information – for example, if it would undermine the purpose of the collection, or if it's just not possible to tell them.

4 You may only collect personal information in ways that are lawful, fair and not unreasonably intrusive. Take particular care when collecting personal information from children and young people.

5 You must make sure that there are reasonable security safeguards in place to prevent loss, misuse or disclosure of personal information. This includes limits on employee browsing of other people's information.

6 People have a right to ask you for access to their personal information. In most cases you have to promptly give them their information. Sometimes you may have good reasons to refuse access. For example, if releasing the information could:

- endanger someone's safety
- create a significant likelihood of serious harassment
- prevent the detection or investigation of a crime
- breach someone else's privacy.

Privacy Commissioner
Te Mana Mātāpono Matatapu

7 A person has a right to ask an organisation or business to correct their information if they think it is wrong. Even if you don't agree that it needs correcting, you must take reasonable steps to attach a statement of correction to the information to show the person's view.

8 Before using or disclosing personal information, you must take reasonable steps to check it is accurate, complete, relevant, up to date and not misleading.

9 You must not keep personal information for longer than is necessary.

10 You can generally only use personal information for the purpose you collected it. You may use it in ways that are directly related to the original purpose, or you may use it another way if the person gives you permission, or in other limited circumstances.

11 You may only disclose personal information in limited circumstances. For example, if:

- disclosure is one of the purposes for which you got the information
- the person concerned authorised the disclosure
- the information will be used in an anonymous way
- disclosure is necessary to avoid endangering someone's health or safety
- disclosure is necessary to avoid a prejudice to the maintenance of the law

12 You can only send personal information to someone overseas if the information will be adequately protected. For example:

- the receiving person is subject to the New Zealand Privacy Act because they do business in New Zealand
- the information is going to a place with comparable privacy safeguards to New Zealand
- the receiving person has agreed to adequately protect the information – through model contract clauses, etc.

If there aren't adequate protections in place, you can only send personal information overseas if the individual concerned gives you express permission, unless the purpose is to uphold or enforce the law or to avoid endangering someone's health or safety.

13 A unique identifier is a number or code that identifies a person in your dealings with them, such as an IRD or driver's licence number. You can only assign your own unique identifier to individuals where it is necessary for operational functions. Generally, you may not assign the same identifier as used by another organisation. If you assign a unique identifier to people, you must make sure that the risk of misuse (such as identity theft) is minimised.

Content updated January 2021. Design updated March 2025.

Privacy Commissioner
Te Mana Mātāpono Matatapu

https://www.privacy.org.nz/assets/Privacy-Act-2020-content/20250326-A-quick-tour-of-the-privacy-principles.pdf

Version as at 1 May 2026

Privacy Act 2020

Public Act 2020 No 31
Date of assent 30 June 2020
Commencement see section 2

Tutorials this week

Codes of Ethics

Engineering NZ

1. Take reasonable steps to safeguard health and safety
2. Have regard to effects on environment
3. Report adverse consequences
4. Act competently
5. Behave appropriately
6. Inform others of consequences of not following advice
7. Maintain confidentiality
8. Report breach of Code



IEEE Code of Ethics

1. to accept responsibility in making decisions consistent with the safety, health, and welfare of the public, and to disclose promptly factors that might endanger the public or the environment;
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
3. to be honest and realistic in stating claims or estimates based on available data;
4. to reject bribery in all its forms;
5. to improve the understanding of technology; its appropriate application, and potential consequences;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others;
8. to treat fairly all persons and to not engage in acts of discrimination based on race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;
9. to avoid injuring others, their property, reputation, or employment by false or malicious action;
10. to assist colleagues and co-workers in their professional development and to support them in following this code of ethics.

ACM Code of Ethics

1. General Moral Imperatives

- 1.1 Contribute to society and human well-being.
- 1.2 Avoid harm to others.
- 1.3 Be honest and trustworthy.
- 1.4 Be fair and take action not to discriminate.
- 1.5 Honor property rights including copyrights and patent.
- 1.6 Give proper credit for intellectual property.
- 1.7 Respect the privacy of others.
- 1.8 Honor confidentiality.

- ...

http://www.acm.org/about/code-of-ethics

Case: Megaupload

RNZ News Life Radio Podcasts Video Sections

RNZ National LIVE
Māpuna

World Politics Crime / Justice Business Sport Health Education Environment Community Country Weather Te Ao Māori Pacific IndoNZ 中文

News / Science and technology 1 July 2026, 8:21pm

Copy URL

Court of Appeal dismisses Kim Dotcom challenge to US extradition

Catherine Hutton for NZ Herald



Related stories

Kim Dotcom timeline: An extradition 13 years in the making
16 August 2024



Kim Dotcom extradition: Lawyer attacks 'grossly disproportionate' treatment over Megaupload
21 April



Decision to extradite Kim Dotcom ruled legal by High Court
10 September 2025



Kim Dotcom says he is recovering from 'serious stroke' as extradition fight continues
26 November 2024



Kim Dotcom to fight extradition, says he won't get fair trial in US
16 August 2024



Case: Megaupload

Kim Dotcom founded the file-sharing site MegaUpload in 2005. The site lets users share copyrighted (and other) material with one another, (potentially) infringing copyright laws. It was responsible for 4% of internet traffic in 2011.

In Jan 2012 Megaupload was shut down by the US Justice Dept and Dotcom was indicted for running an organisation dedicated to copyright infringement. He was arrested (in an (illegal) armed raid in NZ) in Feb 2012.

The case went through many legal instances. Dotcom lives in Glenorchy awaiting likely expedition to the United States.



Case: Megaupload

In Feb 2015, Andrus Nõmm (one of MegaUpload’s programmers) was sentenced to a year’s imprisonment in the US for copyright infringement and racketeering.



Source: www.youtube.com/watch?v=TC6YzYu1vd0



(Brief) In-Class Activity



Megaupload Debate

We will divide up the class into two groups A and B.
Group A should debate in favour of Andrus Nõmm.
Group B should debate against him.

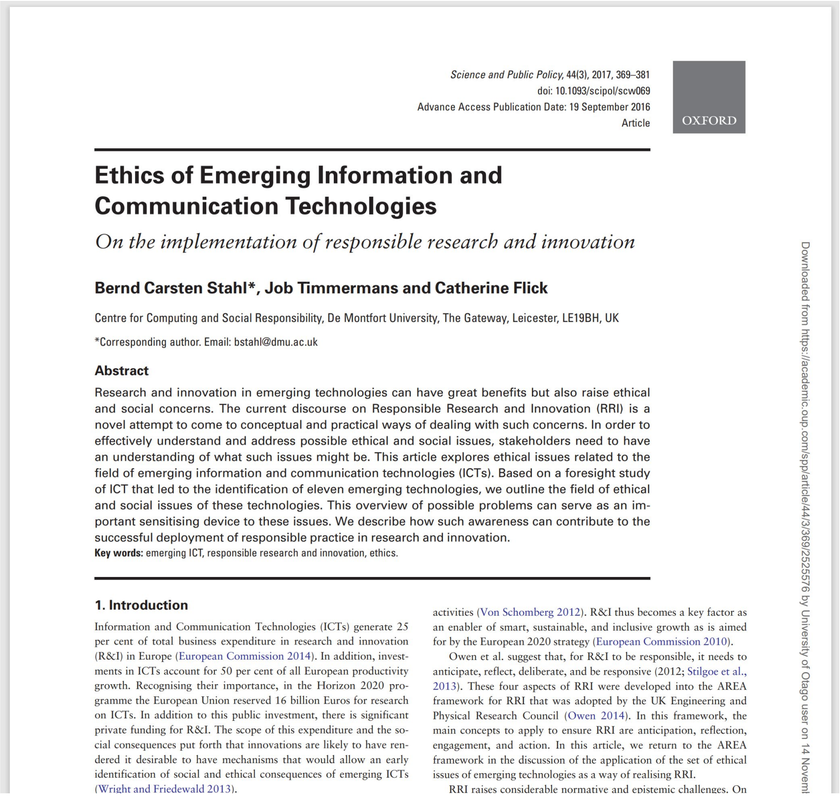
- What are the conflicts in a file-sharing case like this one?
- What position should governments take on file-sharing sites?
- Should programmers be held accountable for what is done with their work?
- What (other) ethical questions arise and how should they be addressed?

Ethics and Emerging Technologies

Guiding Questions for Ethics of Emerging ICT

(derived from Stahl, Timmermans, and Flick (2017)

- Impact on Individuals
- Consequences on Society
- Uncertainty of outcomes
- Perceptions of technology
- Role of humans



Science and Public Policy, 44(3), 2017, 369–381
doi: 10.1093/scipol/scw069
Advance Access Publication Date: 19 September 2016
Article

OXFORD

Ethics of Emerging Information and Communication Technologies

On the implementation of responsible research and innovation

Bernd Carsten Stahl*, Job Timmermans and Catherine Flick

Centre for Computing and Social Responsibility, De Montfort University, The Gateway, Leicester, LE19BH, UK

*Corresponding author. Email: bstahl@dmu.ac.uk

Abstract

Research and innovation in emerging technologies can have great benefits but also raise ethical and social concerns. The current discourse on Responsible Research and Innovation (RRI) is a novel attempt to come to conceptual and practical ways of dealing with such concerns. In order to effectively understand and address possible ethical and social issues, stakeholders need to have an understanding of what such issues might be. This article explores ethical issues related to the field of emerging information and communication technologies (ICTs). Based on a foresight study of ICT that led to the identification of eleven emerging technologies, we outline the field of ethical and social issues of these technologies. This overview of possible problems can serve as an important sensitising device to these issues. We describe how such awareness can contribute to the successful deployment of responsible practice in research and innovation.

Key words: emerging ICT, responsible research and innovation, ethics.

1. Introduction

Information and Communication Technologies (ICTs) generate 25 per cent of total business expenditure in research and innovation (R&I) in Europe (European Commission 2014). In addition, investments in ICTs account for 50 per cent of all European productivity growth. Recognising their importance, in the Horizon 2020 programme the European Union reserved 16 billion Euros for research on ICTs. In addition to this public investment, there is significant private funding for R&I. The scope of this expenditure and the social consequences put forth that innovations are likely to have rendered it desirable to have mechanisms that would allow an early identification of social and ethical consequences of emerging ICTs (Wright and Friedewald 2013).

activities (Von Schomberg 2012). R&I thus becomes a key factor as an enabler of smart, sustainable, and inclusive growth as is aimed for by the European 2020 strategy (European Commission 2010).

Owen et al. suggest that, for R&I to be responsible, it needs to anticipate, reflect, deliberate, and be responsive (2012; Stilgoe et al., 2013). These four aspects of RRI were developed into the AREA framework for RRI that was adopted by the UK Engineering and Physical Research Council (Owen 2014). In this framework, the main concepts to apply to ensure RRI are anticipation, reflection, engagement, and action. In this article, we return to the AREA framework in the discussion of the application of the set of ethical issues of emerging technologies as a way of realising RRI.

RRI raises considerable normative and epistemic challenges. On

Downloaded from https://academic.oup.com/scipol/article/44/3/369/2525576 by University of Otago user on 14 Novem

Impact on Individuals

Privacy

- Which types and quantities of data will the Emerging Technology require and/or generate?
- Who will have access to the data?
- Who will know about the existence and possible inferences from the data?

Autonomy

- In what ways does the Emerging Technology improve autonomy of the user?
- To what extent does the Emerging Technology monitor or control people's behaviour, attitude, emotions, thoughts, moods, and actions?
- Does the Emerging Technology make decisions? What are these decisions based on, and do they take into account ethical issues?

Treatment of Humans

- In what ways could the Emerging Technology impact the daily life of users?
- Could vulnerable users be particularly affected by the Emerging Technology?
- Does the Emerging Technology seek informed consent where necessary?
- Could events that happen within the virtual world of the technology negatively impact on the real world?

Impact on Individuals

Identity (Could it alter the user's view on what it means to be a human)

- Does the Emerging Technology change human capabilities, e.g. their ability to perform certain tasks?
- How will the Emerging Technology affect the way in which users see themselves or one another?

Security

- Is the Emerging Technology likely to create novel types of vulnerabilities, e.g. by generating or requiring sensitive data?
- To what degree will existing security solutions be applicable to the Emerging Technology?

Consequences on Society

Digital Divides

- Which impact will the Emerging Technology have on the possibility to participate in social life?
- Which mechanisms of diffusion are likely to be used to introduce the Emerging Technology widely?
- What are the likely consequences for groups that are already marginalised?

Collective human identity and the good life

- Does the Emerging Technology replace established human activities or work?
- Which view of culture or human society is the Emerging Technology likely to promote?

Ownership, data control, and intellectual property

- Which ownership or access models are favoured by the Emerging Technology?
- Does the Emerging Technology make use of open or proprietary standards?

Consequences on Society

Responsibility

- Which existing and legacy system does the Emerging Technology rely on?
- Who is responsible for testing of the Emerging Technology systems?
- Which consequences could a malfunction or misuse of the Emerging Technology have?

Surveillance

- Will the data that the Emerging Technology generates allow for surveillance?
- How are access rights embedded in the Emerging Technology?

Cultural differences

- Which assumptions about normal and desirable behaviour are embedded in the Emerging Technology?
- Is there a possibility of testing the Emerging Technology in different cultures?

Uncertainties, Perceptions, Humans

Uncertainty of outcomes

- What are the possible uses of the Emerging Technology beyond the ones primarily envisaged?
- Are there foreseeable side effects or unintended consequences of the Emerging Technology?

Perceptions of technology

- Will the Emerging Technology appear autonomous to users?
- Will the Emerging Technology be anthropomorphic, that is, look or act in ways that we normally expect humans to look or behave?
- Which human activities will be replaced by the Emerging Technology?

Role of humans (change the way in which we see ourselves individually and collectively and the way we can interact.)

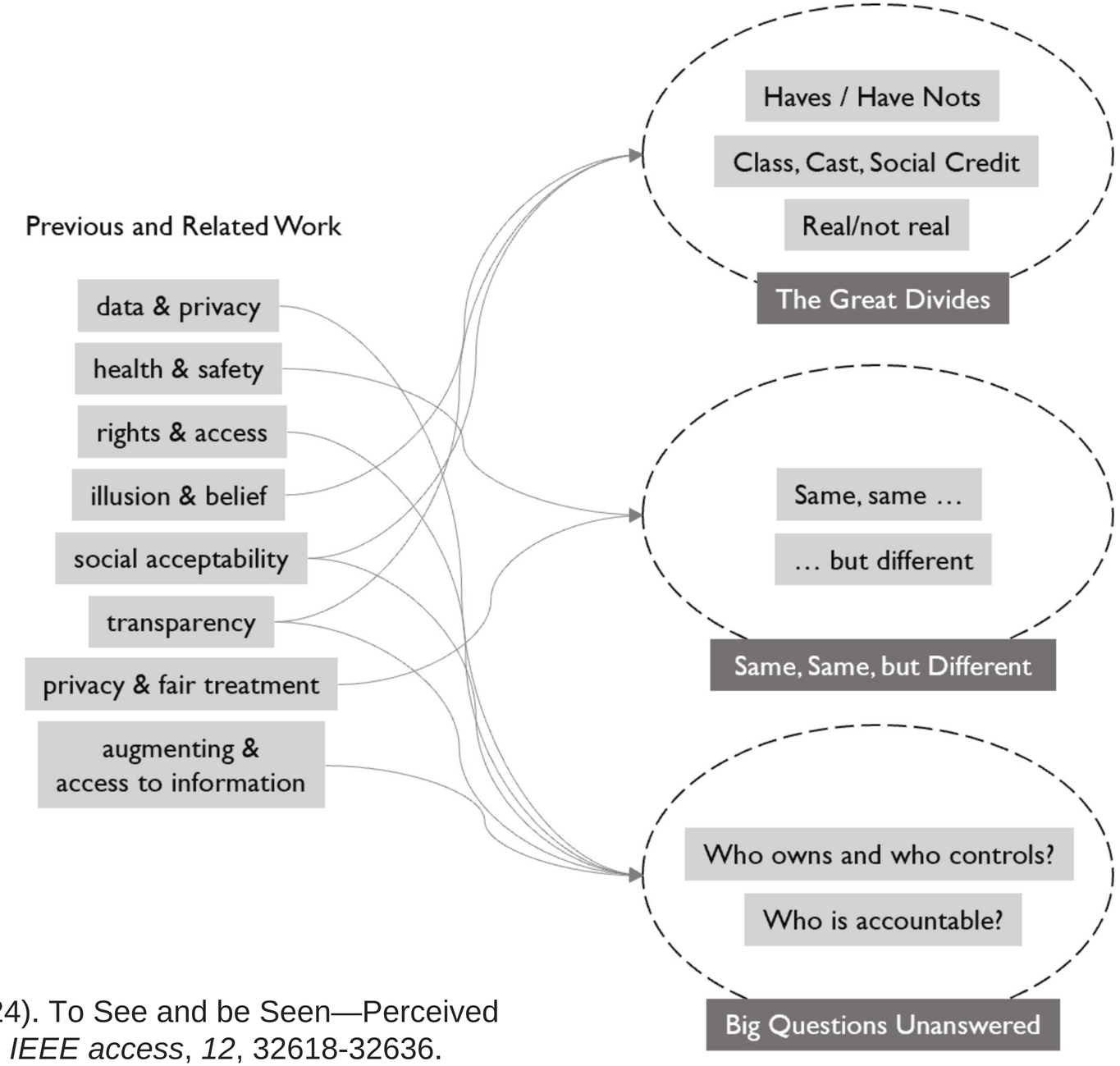
- Which novel capabilities will the Emerging Technology provide users with?
- Will the Emerging Technology be closely linked to the user (e.g. be wearable) or implanted?
- Is the Emerging Technology likely to replace established human activities or work?

Example

Pervasive Augmented Reality as an emerging technology

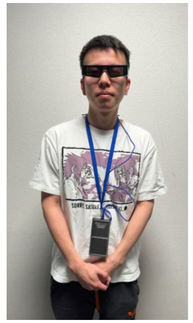
PAR Ethics Studies

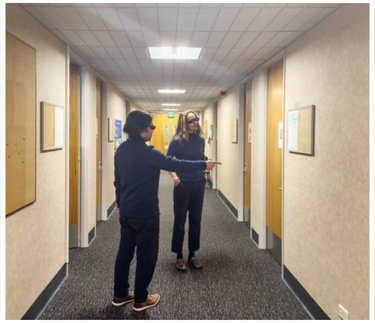


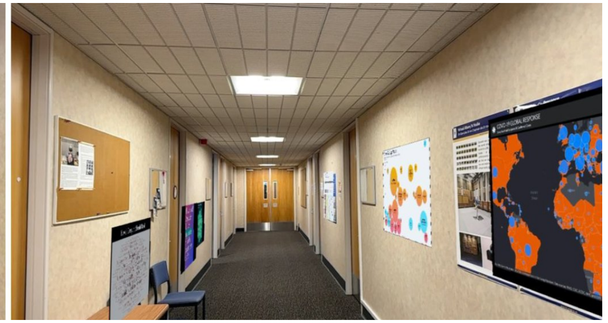


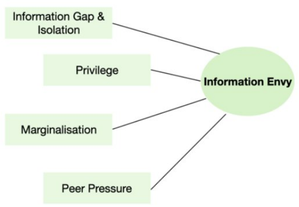
Regenbrecht, H., Knott, A., Ferreira, J., & Pantidi, N. (2024). To See and be Seen—Perceived Ethics and Acceptability of Pervasive Augmented Reality. *IEEE access*, *12*, 32618-32636.

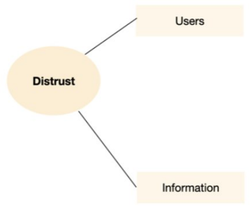
PAR Ethics Studies

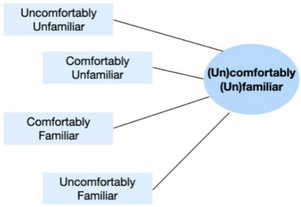


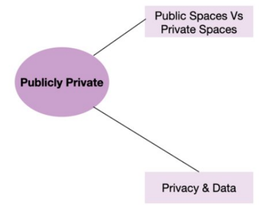




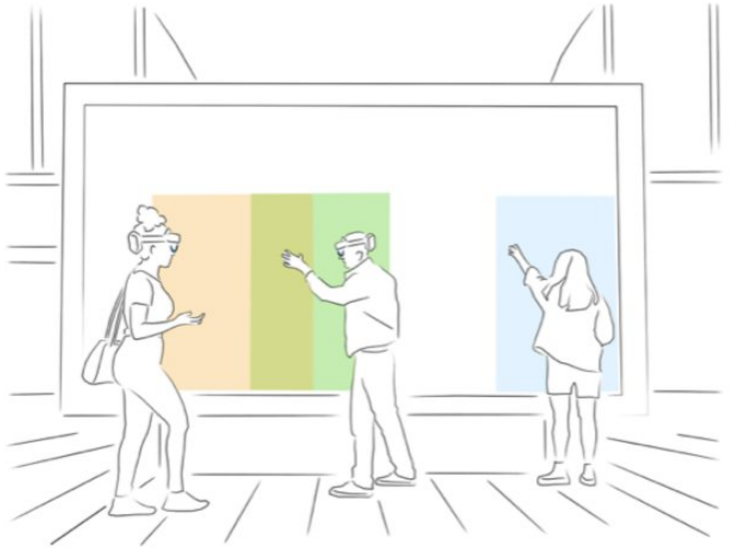








Motivation



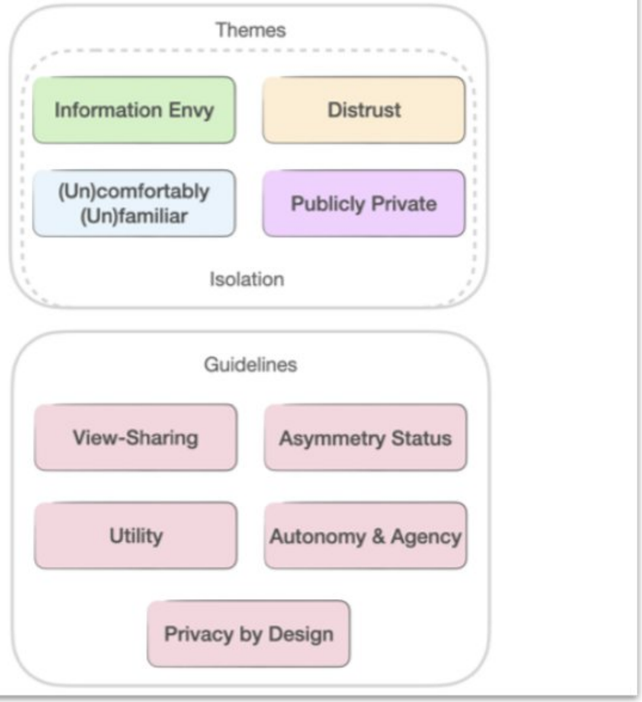
- Ambiguous ownership of space and content when using Pervasive AR for Public Displays.

Study



- Pair study (n=40)
- Semi public
- Two conditions
  - Information coherence
  - Priming description
- Qualitative and quantitative measurements
- Technology probe
- Thematic analysis
- Development of guidelines

Findings



Perera, K., Langlotz, T. Pantidi, N., and Regenbrecht, H. (2024).
What you see is (not necessarily) what I see—Pervasive AR for Public Displays. *Proceedings of OzCHI 2024.*

PAR Ethics Studies

Condition 1: Symmetric-Aware



Jason's view Sarah's view

Condition 2: Asymmetric-Aware

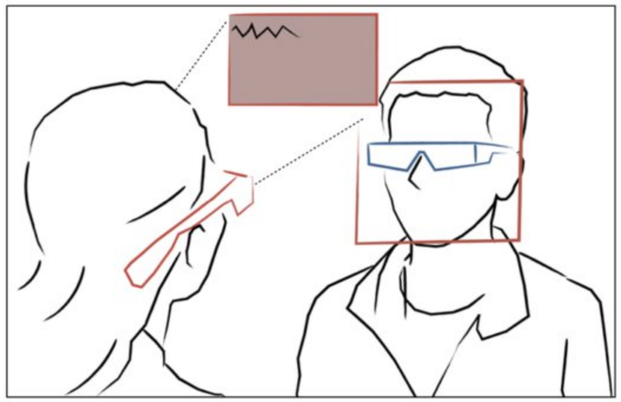


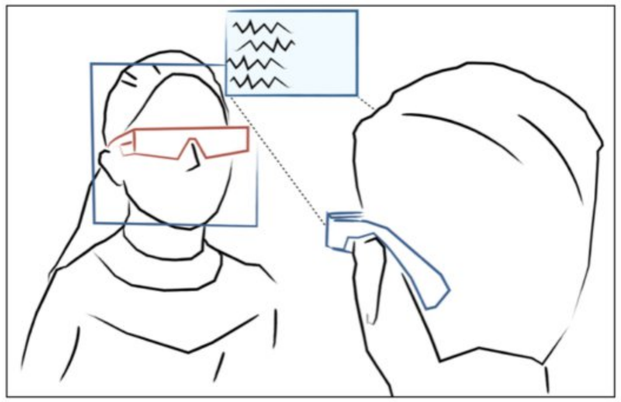
Jason's view Sarah's view

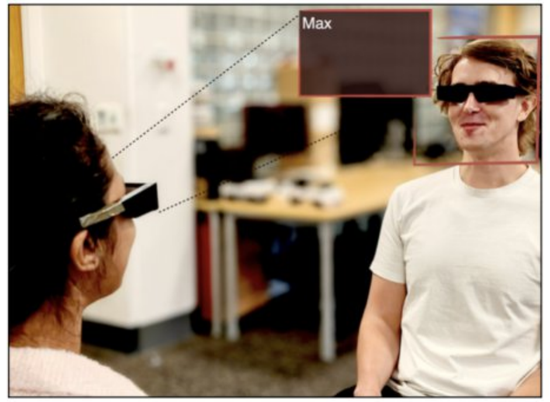
Condition 3: Asymmetric-Unaware

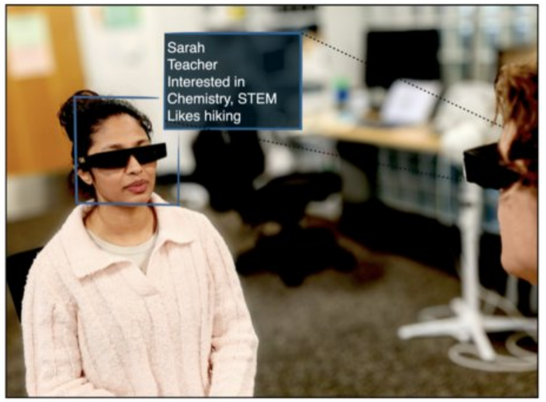


Jason's view Sarah's view





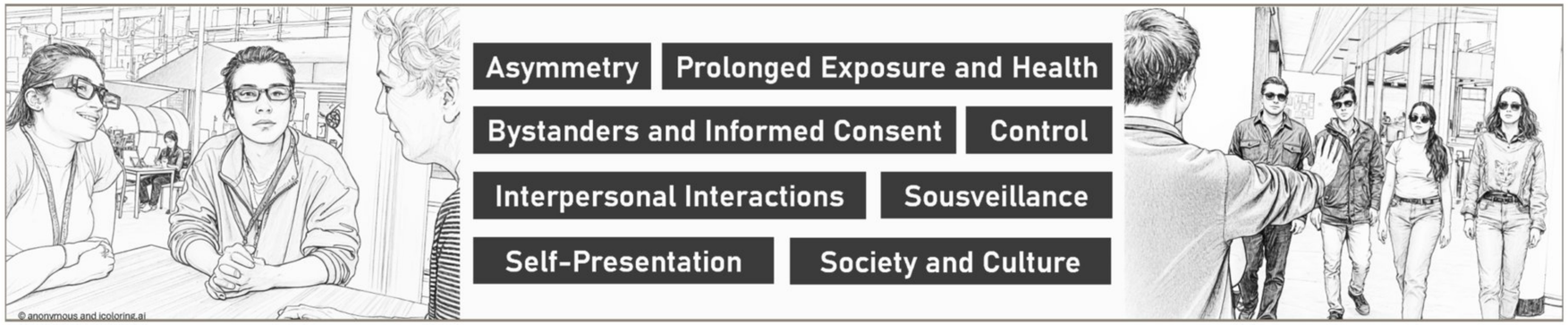




- First Impression Formation
  - Information-based First Impressions
  - Information-based Self-Presentation
  - Asymmetry-based First Impressions
- Changing Social Behaviour
  - Initiating Interactions
  - Unpleasant Social Situations
- Modulating Interactions
  - Favourable Modulations
  - Unfavourable Modulations
- Negotiating Control
  - What Information?
  - How much Information?
  - Whose Information?
  - How is it triggered?
- User Agency
  - Power Imbalance
  - Symmetry vs Asymmetry
  - Awareness
- Conditional Acceptability
  - Factual vs Personal Information
  - Specific Contexts

Perera, K., Langlotz, T. Pantidi, N., and Regenbrecht, H. (2025).
Don't look at me like that—How AR face recognition changes our social behaviour. Proceedings ISMAR 2025

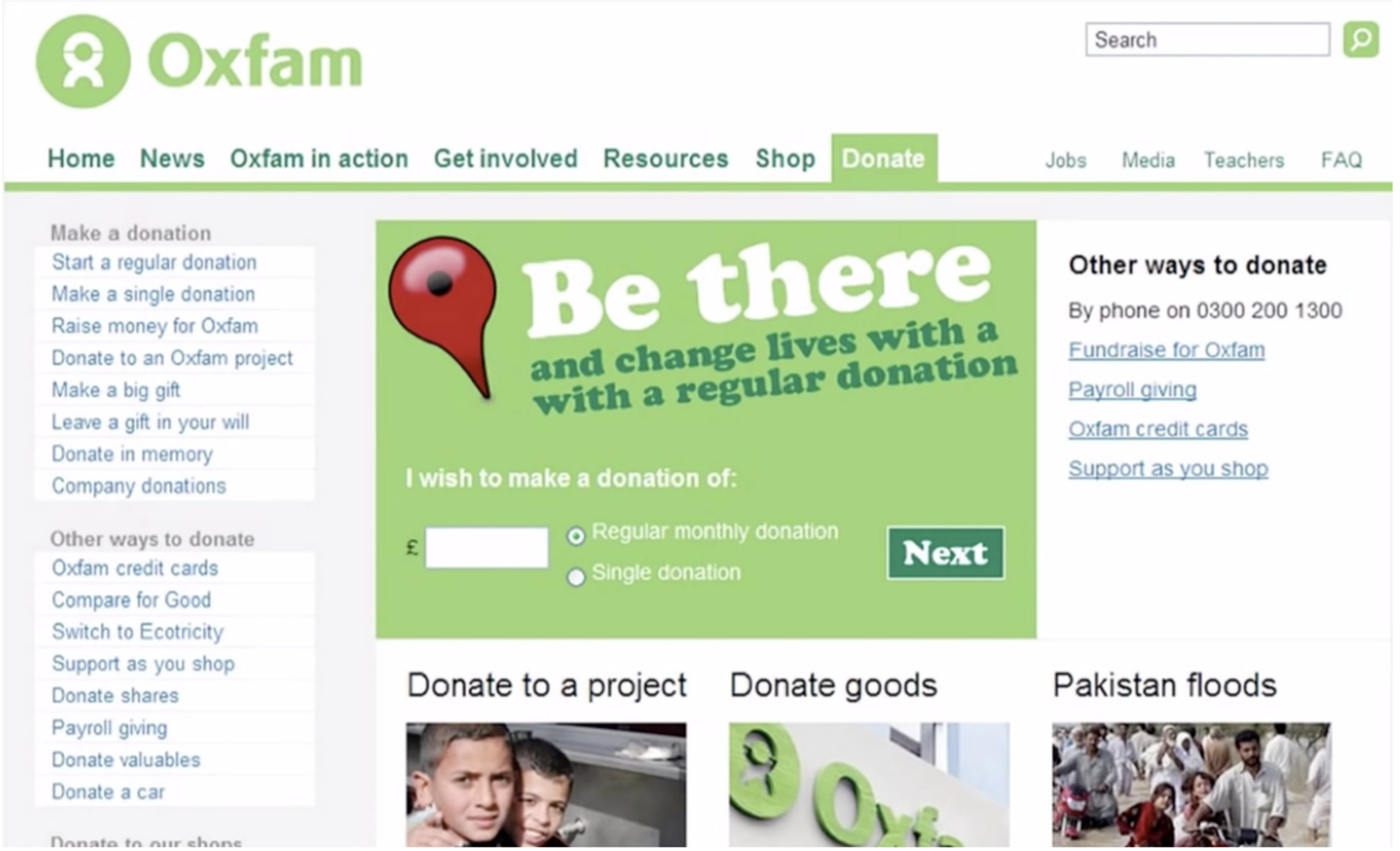
Lab Task Friday



User Interface Dark Patterns

Dark Patterns of UI

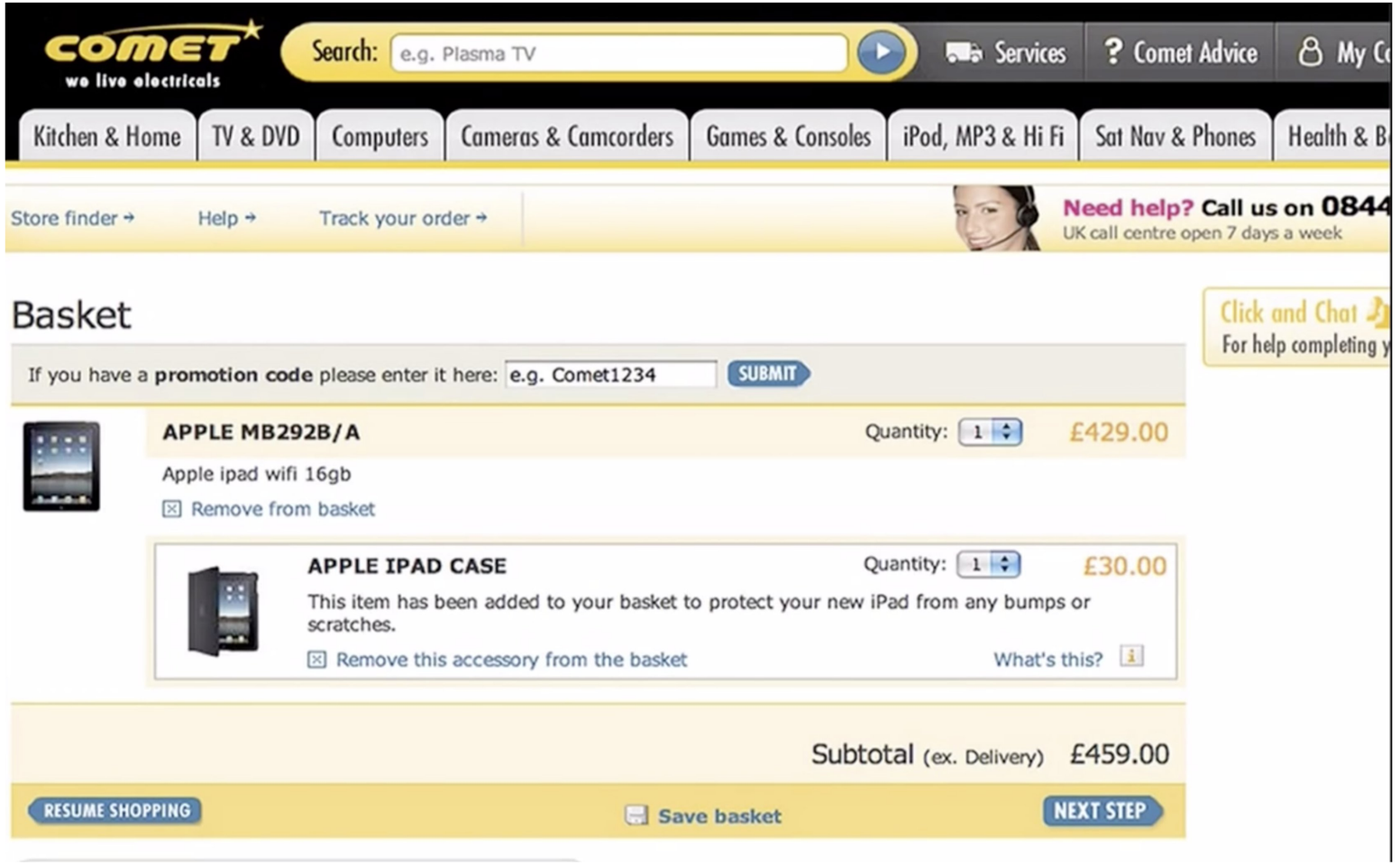
Turn design patterns against the user - ethical?



Some examples from: “Dark Patterns: User Interfaces Designed to Trick People”

Dark Patterns of UI

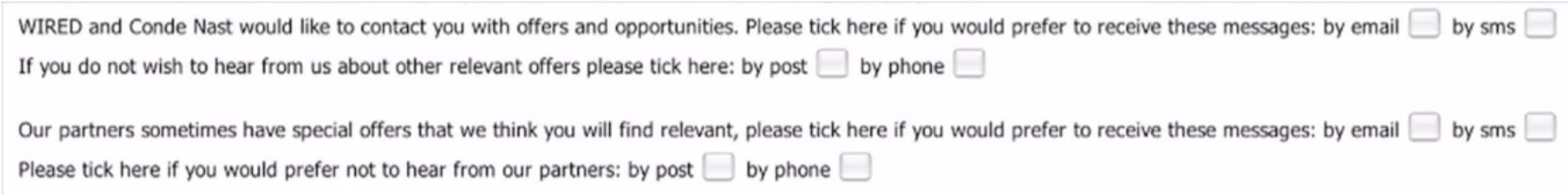
Turn design patterns against the user - ethical?



Some examples from: “Dark Patterns: User Interfaces Designed to Trick People”

Dark Patterns of UI

Turn design patterns against the user - ethical?



Some examples from: “Dark Patterns: User Interfaces Designed to Trick People”

Dark Patterns of UI

Turn design patterns against the user - ethical?



Some examples from: “Dark Patterns: User Interfaces Designed to Trick People”

Dark Patterns of UI

Turn design patterns against the user - ethical?



Some examples from: “Dark Patterns: User Interfaces Designed to Trick People”

Dark Patterns of UI

INDEPENDENT

Subscribe LOGIN

NEWS INDEPENDENT TV CLIMATE SPORT VOICES CULTURE PREMIUM INDY/LIFE INDYBEST INDY100 MY INDEPENDENT VOU

Tech

Amazon intentionally made it difficult to cancel Prime subscription in secret project

Project 'Iliad' put multiple layers of questions and offers between the main Amazon page and the final cancellation

Adam Smith • Thursday 17 March 2022 16:26 • Comments



TechRadar is supported by its audience. When you purchase through links on our site, we may earn an affili

Home > News > Internet

Amazon Prime memberships are now harder to cancel – and it's no accident

By Gerald Lynch last updated 19 days ago

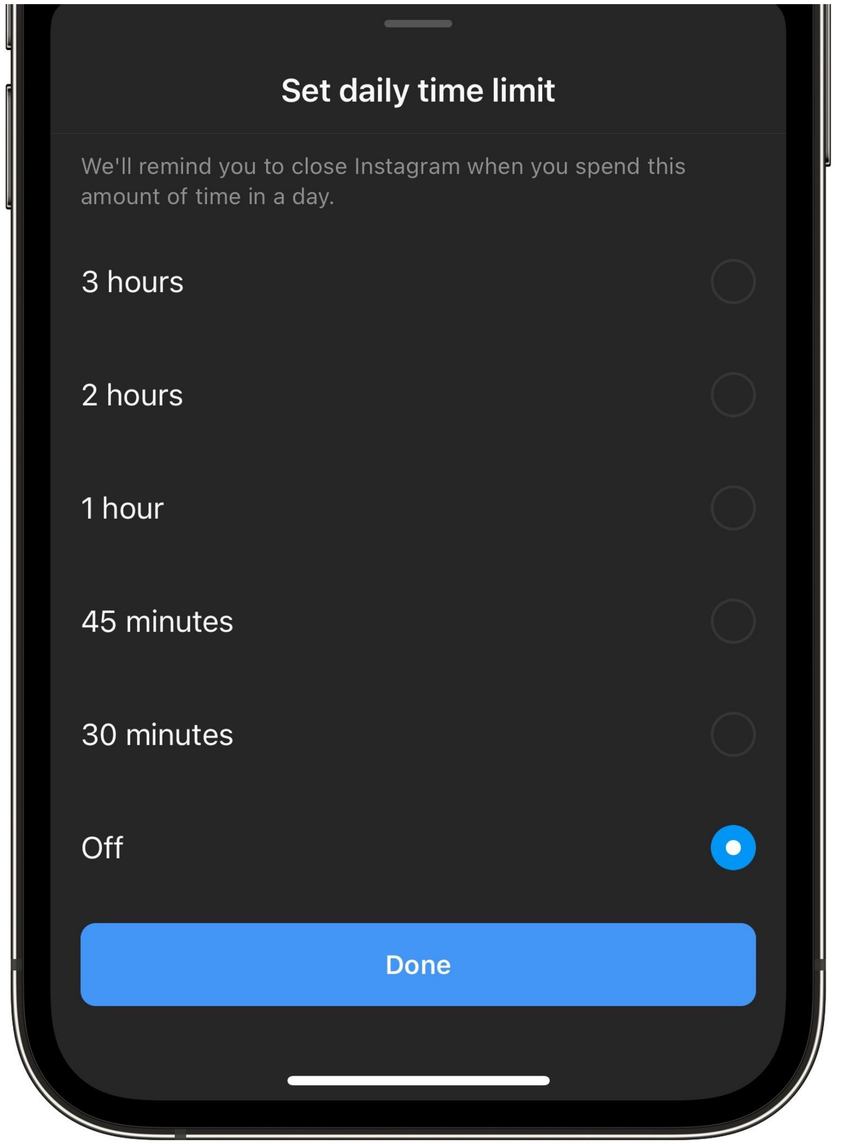
Amazon accused of using 'dark patterns'



(Image credit: Shutterstock)

Dark Patterns of UI





Instagram



(Brief) In-Class Activity



If you are not paying for the product, you are the product.

Andrew Lewis

With your neighbours in class develop a search engine UI (Google-style, ChatGPT-style, Whatever-style) which uses Dark Patterns of UI or other (deceiving) techniques or strategies to exploit your users. Try to stay within what's legally acceptable (if you know it).*



Guy Abramow

* this exercise is intended to make you think about consequences of UI design, not to apply those in professional practice←

Summary

- Ethics - Morality
- Personal and Professional Co-Responsibility
- Privacy (see also Tutorial)
- Codes of Ethics
- Guiding Questions for Emerging Technologies (see also Lab)
- Dark Patterns

The end!

www.hci.otago.ac.nz